Assignment No 5

Q1 Count Digits in a Number

Given a number n, count how many digit it contains.

n = 54321
o/p = 5

code :-

```
int n = 54321
int count = 0;
while ( n != 0) {
    int rem = n%10;
    count ++;
    n /= 10;
}
sop (count);
```

dry run

Iteration	n (before)	rem = n%10	count	n = n/10 (after)
1	54321	1	1	5432
2	5432	2	2	543
3	543	3	3	54
4	54	4	4	5
5	5	5	5	0

first iteration

n = 54321;
count = 0;
~~while~~ condition true,
rem = 1
count++ // 0 → 1
n = n/10 → 54321/10 = 5432

2nd

n = 5432
count = 1
condn true
rem = 2
count++ // 1 → 2
n/=10 // 5432/10 = 543

3rd

n = 5432
count = 2
rem = 3
count++; → 2 → 3
n/=10 // 54

4th

n = 54
count = 3;
rem = 4
count++ // 3 → 4
n/=10 // 5

5th iteration

n = 5
count = 4
rem = 5
count++ // 4 → 5
n/=10 // 0

finally आपल्याला सगळे iteration complete झाल्यावर count 5 मिळला.

Q. 2 Extract All Digits.

n = ~~5432~~ 5382

o/p 2
8
3
5

code :-

```
int n = 5382
while (n != 0) {
    int rem = n % 10;
    cop (rem);
    n /= 10;
}
```

dryrun

iteration 1
n = 5832
condition true.
rem = 2
print (2)
n = 583

2nd iteration

n = 583
condn true
rem = 3
print 3
n = 58

3rd iteration
n = 58
condn true
rem = 8
print 8
n = 5

4th iteration
n = 5
condn true
rem = 5
print 5
n = 0

5th iteration
n = 5
condn false
exit loop.

Q3. Reverse a Number

I/p = 12345

O/p = 54321

code :-

```
int n = 12345;
int rev = 0;
while( n != 0){
    int rem = n % 10;
    rev = rev*10 + rem;
    n /= 10;
}
sop(rev);
}
```

Dry run

1st iteration

n = 12345

rev = 0

cond^n true

rem = 5

rev = 0*10+5 = 5

n = 1234

2nd iteration

n = 1234

rev = 5

cond^n true

rem = 4

rev = 5*10+4 = 54

n = 123

3rd iteration

n = 123

rev = 54

cond^n true

rem = 3

rev = 54*10+3 = 543

n = 12

4th iteration

n = 12

rev = 543

cond^n true

rem = 2

rev = 543*10+2 = 5432

n = 1

5th iteration
n = 1
rev = 5432
condⁿ true
rem = 1
rev = 5432*10+1 = 54321
n = 0

फिर शेवटी आपण rev हे variable print करू

Q.4 Palindrome Number
Input : 121
O/p - palindrome
I/p = 123
O/p not palindrome

```
int n = 121
int original = n;
int rev = 0;
while( n != 0 ){
    int rem = n % 10;
    rev = rev*10 + rem;
    n /= 10;
}
if (original == rev) {
        sop ("palindrome").
else{
        sop ("Not palindrome");
}
```

1st iteration
n = 121
~~cond~~ rev = 0
condn true
rem = 1
rev = rev*10+rem
= 0*10+1 = 1
n = 12

2nd iteration
n = 12
rev = 1
condn true
rem 2
rev = 12
n = 1

3rd iteration
n = 1
rev = 12
cond true
rem = 1
rev = 121
n = 0

सगळे iteration complete झाल्यावर if ची condition check करणार -

if (original == rev)

तर ही condn true झाली तर Palindrom आहे नाहीतर palindrome नाही.

Q.5 Sum of digits
I/p = 543 O/p = 12

```
int n = 543
int sum = 0;
    while (n != 0) {
        int rem = n % 10;
        sum = sum + rem
        n /= 10;
    }
    Sop (sum);
}
```

1st iteration
n = 543
sum = 0
cond^n true
rem = 3
sum = sum+rem;
// = 0 + 3
= 3
n = 54

2nd iteration
n = 54
sum = 3
cond^n true
rem = 4
sum = sum + rem
= 3 + 4
= 7
n = 5

3rd iteration
n = 5
sum = 7
cond^n true
rem = 5
sum = sum + rem;
= 7 + 5
= 12
n = 0;

ત્યારે n નું 0 થતાં ત્યારે while loop ની બહાર નીકળી અને sum print કરાવી.

Q.8

Q.7 Print All Divisor

I/p = 12 O/p : 1 2 3 4 6 12

```
int n = 12

for (int i=1; i<=n; i++){
    if (n%i == 0) {
        cop(i);
    }
```

loop i	Condn $i \le n$	condn if $n \% i == 0$	o/p	increment
1	$1 \le 12$ ✓	✓	1	i++
2	~~2~~ ✓	✓	2	i++
3	✓	✓	3	i++
4	✓	✓	4	i++
5	✓	×		i++
6	✓	✓	6	i++
7	✓	×		i++
8	✓	×		i++
9	✓	×		i++
10	✓	×		i++
11	✓	×		i++
12	✓	✓	12	i++

final o/p 1 2 3 4 6 12

Q.8 Count ~~of~~ Number of Divisors

$n = 12$

Divisors : 1 2 3 4 6 12

~~count~~ o/p = 6

```
int n=12
for (int i=1; i<=n; i++) {
    if (n%i == 0) {
        cout++;
    }
}
cout (count);
```

loop i	cond^n $i \le n$	cond^n if $(n\%i==0)$	count
	✓	✓	1
1	✓	✓	2
2	✓	✓	3
3	✓	✓	4
4	✓	✗	
5	✓	✓	5
6	✓	✗	
7	✓	✗	
8	✓	✗	
9	✓	✗	
10	✓	✗	
11	✓	✗	
12	✓	✓	6

Q9 Check Prime Number.

I/p = 7 o/p: prime

I/p = 9 o/p: not prime

```
int n = 9;
int count = 0;
for (int i = 1; i ≤ n; i++) {
        if (n%i == 0) {
            count++;
        } else {
    }
    if(count == 2) {
        Sop ("prime");
    } else {
        Sop ("not prime");
    }
}
```

loop i	condⁿ i≤n	if (n%i==0)	count
1	✓	✓	1
2	✓	×	
3	✓	×	
4	✓	×	
5	✓	×	
6	✓	×	
7	✓	✓	2

शेवटी count print करतो.
जर if (count == 2) असेल तर तर
~~count~~ तो number prime असतो.

Q.6 Amstrong Number

~~int original~~
Input = 153

```
int num = 153;
int original = num;
int sum = 0;
int counts = 0;
~~int temp = num~~;
while (num != 0) {
    int digit = num % 10;

    int num = digit * digit * digit
    sum += num;
    num /= 10;
}
if (original == sum) {
    sop ("amstrong number");
} else {
    sop ("not amstrong number.
```

Dry run.

① num = 153
digit = 153 % 10 = 3
num = 3 * 3 * 3 = 27
sum = 0 + 27 = 27
153 / 10 = 15

② num = 15
digit = 15 % 10 = 5
num = 5 * 5 * 5 = 12
sum = 27 + 125
= 152
15 / 10 = 1

③ num=1
digit = 1%10 = 1
n = 1*1*1 = 1
sum = 152+1 = 153
n/10 = 1/10 = 0

while condⁿ get false
then code goes to if else statemen

Q.11 GCD / HCF of two numbers.
Input
a = 2
b = 18
o/p = 6

Code:
```
int a = 12
int b = 18
int gcd = 1

for (int i=1; i<=(Math.min(a,b)); i++) {

    if (a%i==0 && b%i==0) {
        gcd = i;
    }
}
sop (gcd);
```

Dry run

i=1 ; i<=12 (True)
if (12%1==0 && 18%1==0) (True)
gcd=1;

i=2 ; i<=2 (True)
if (12%2==0; && 18%2==0) True
gcd = 2;

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i = 3 3 <= 12 (True)
if (12 % 3 == 0 && 18 % 3 == 0) (True)
gcd = 3

i = 4 4 <= 12 (True)
if (12 % 4 == 0 && 18 % 4 == 0) false

i = 5 5 <= 12 (True)
if (12 % 5 == 0 && 18 % 5 == 0) (false)

i = 6 6 <= 12 (True)
if (12 % 6 == 0 && 18 % 6 == 0) True
gcd = 6;

i = 7 7 <= 12 (True)
if (12 % 7 == 0 && 18 % 7 == 0) (false)

i = 8 8 <= 12 (True)
if (12 % 8 == 0 && 18 % 8 == 0) false

i = 9 9 <= 12 True
if (12 % 9 == 0 && 18 % 9 == 0) (false)

i = 10 10 <= 12 (True)
if (12 % 10 == 0 && 18 % 10 == 0) false

i = 11 11 <= 12 (True)
if (12 % 11 == 0 && 18 % 11 == 0) false

~~i == 12~~

i = 12 12 <= 12 (True)
if (12 % 12 == 0 && 18 % 12 == 0) false

12 GCD using Euclidean Algorithm

Input

a = 48

b = 18

O/p = 6

Code

```
int a = 48;
int b = 18;
int gcd = 0;
while (a != 0 && b != 0) {
    if (a>b) {
        a = a % b;
    } else {
        b = b % a;
    }
    if (a == 0) {
        gcd = b
    } else {
        gcd = a
    }
}
sop (gcd);
```

Dry run

① a = 48 ; b = 18 ;
while (a != 0 && b != 0) (true)
if (a>b) true
a = 48 % 18 = 12
}
if (a==0){ //false
}else{
gcd = 12 ;

② a = 12 , b = 18
while () true
if (a>b) false
}else {
b = 18 % 12 = 6
}
if (a==0) { //false
}else{
gcd = 12
}

③ a = 12 b = 6
while () //true
if (a>b) true
a = 12 % 6 = 0
;
if (a==0) true
gcd = 6 ;

④ a = 0 b = 6
while ()
false

शेवटी while ची condition तेव्हा fail होईल तेव्हा आपली gcd ही 6 असेल.

24) Check Perfect Number

Input = 6
Divisor 1,2,3
O/P : perfect Number.

```
int n = 6
int sum = 0;

for(int i=1; i<=n/2; i++) {
    if (n%i==0) {
        sum=sum+i;
    }
}
if (n==sum) {
    sop("perfect number");
}else{
    sop("not a perfect number");
}
```

dry run
आपला आपला loop हे n/2 पर्यंत फिरवतो.

~~i=1~~ n = 6 n/2 = 3

i = 1 i ≤ 3 (True)
if (n%i == 0) // True
sum = 0+1 sum = 1

i = 2 2 ≤ 3 (True
if (6%2 == 0) True.
sum = 1 + 2 = 3

True
i = 3 3 ≤ 3
if (6%3 == 0) -
sum = 3 + 3
= 6
शेवटी जर आपली आतली sum आणि n जर equal असतील तर तो Perfect.

13. Check Coprime Number
Input :
8 15
output
coprime

```
int a = 15;
int b = 8;
int gcd = 0;

for (int i = 1; i <= Math.min(a, b) ; i++) {
    if (a%i == 0 && b%i == 0) {
        gcd = i
    }
}
if (gcd == 0) {
    sop ("coprime");
}else {
    sop("not a coprime");
}
```

Dry run
math.min ने ~~आपला~~ minimum value ~~i = +~~ घेते. तर आपला for loop त्यावर फिरतो.

i = 1 1 ≤ 8 True
if (15%1 == 0 && 8%i == 0) True
gcd = 1;

i = 2 i ≤ 2 True
if (15%2 == 0 && 8%2 == 0) false

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$i = 3$ $\quad 3 \leq 8$
if (15%3 == 0 && 8%3 == 0) { false

$i = 4$ $\quad 4 \leq 8$ $\quad$ True
if (15%4 == 0 && 8%4 == 0) false

$i = 5$ $\quad 5 \leq 8$ $\quad$ True
if (15%5 == 0 && 8%5 == 0) false.

$i = 6$ $\quad 6 \leq 8$ $\quad$ True
if (15%6 == 0 && 8%6 == 0) false

$i = 7$ $\quad 7 \leq 8$ $\quad$ True
if (15%7 == 0 && 8%7 == 0) false

$i = 8$ $\quad 8 \leq 8$ $\quad$ True
if(15%8 == 0 && 8%8 == 0) false.